

CLOUD BASED ANALYTICS

FINAL PROJECT

ANALYZE TEACHING ASSISTANT HELP SESSIONS WITHIN KELLY

INSIGHTS DERIVED FROM VISUALIZATIONS

SHEET 1 - APPOINTMENT AVAILABILITY:

We are analyzing the distribution of appointments throughout days of the week and how they correspond to project deadlines, along with the most frequently scheduled slots set up by TA's.

1. **Navigator:** Navigating using Power BI, the journey exposes insights into Appointment Availability and TA bookings, ending in a full examination of lead times.
2. **Slicer:** In Power BI, the slicer quickly filters data, highlighting essential insights by filtering out all TA names, hence easing analysis and focus.
3. **Stacked column chart:** In Power BI depicts appointment availability by day, contrasting the number of booked and open appointments. For instance, the number of open appointments outnumbers those scheduled/booked appointments, indicating possibilities for scheduling improvement. Analysis shows that Tuesday is the busiest day, with the most appointments, followed by Monday and Thursday. In contrast, Wednesday had the fewest appointments, indicating a lighter workload or lower demand on that day. This distribution identifies probable trends in appointment scheduling behavior across the week.
4. **Matrix:** In the Power BI matrix visualization, the table displays regularly scheduled slots organized by "TimeofAppointment" and "CountofID's," with "open" and "booked" sub-columns. Peak demand is between 12:00 PM and 4:00 PM, with the 3:00 PM session having the most appointments (86). The least popular timeslots, on the other hand, are between 8:00 AM and 9:30 AM, with only 29 appointments recorded at each. This insight identifies possible possibilities to improve scheduling efficiency during peak and off-peak hours.
5. **Card:** The card visualization in Power BI provides critical metrics clearly, including total appointments along with total open and total booked appointment counts. These metrics provide a rapid snapshot of overall appointment activity and the distribution of open and booked statuses, which helps to assess appointment management efficiency.

SHEET 2 - TA'S BOOKED:

We are Evaluating which TAs are most or least chosen, identifying shortcomings and figuring out which day has the highest bookings filtered by month.

1. **Stacked Bar Chart:** The stacked bar chart in Power BI displays the number of appointment IDs by TA name, filtered by status booked or open. Particularly, Kevin and Angela have the highest number of bookings, while Anne has the lowest. The remaining TAs typically follow regular booking patterns. This visualization gives information about individual TA performance and appointment distribution by status.
2. **Slicer:** The slicer quickly filters data, highlighting essential insights by filtering based on months of the year.
3. **Clustered bar chart:** Shows the total number of appointments booked by day of booking, with appointment IDs on the x-axis and days of the week on the y-axis. Monday is the busiest day, with 97 appointments, while Friday has the fewest at 20. The remaining weekdays range from 29 to 50 appointments, indicating a somewhat constant booking trend throughout the week. This analysis reveals the distribution of appointment bookings across different days, showing potential scheduling patterns or preferences.

SHEET 3 - DAY SEGMENT INSIGHTS:

Streamlining an extensive understanding of appointment scheduling patterns, appointment segments, and names of TA.

1. **Stacked bar graph:** Filters appointment data by TA names on the x-axis and appointment segments of the day on the y-axis. This graphic shows which TAs have more appointments in the afternoon, morning, or evening, offering information on their schedule preferences and task distribution throughout the day.
2. **Sunburst chart:** Displays the number of appointments organized by day of the week and time of day (morning, afternoon, evening). This visualization provides a hierarchical perspective, allowing for an intuitive understanding of appointment distribution over multiple time segments and days, assisting with scheduling analysis and optimization.

SHEET 4 – WEEKLY APPOINTMENT INSIGHTS:

Analyzing appointments on a weekly basis, the data can be sorted from highest to lowest appointment counts, as well as the proportion of appointments compared to total appointments for that week. This provides a thorough understanding of weekly appointment distribution, highlighting peak periods and allowing for comparison between weeks.

1. **Pie chart:** The Appointment Count By Week, which is filtered by appointment week number, visually displays appointment counts for each week as well as percentages. It provides a concise summary of appointment distribution across weeks, revealing the relative importance of each week in the dataset.
2. **Line Chart:** Shows the number of appointment IDs filtered by appointment week number and status (open or booked). This dynamic visualization provides insights into the trends of both open and booked appointments throughout several weeks, allowing for a thorough investigation of appointment dynamics.
3. **Area Chart:** The Cumulative Appointments shows appointment counts cumulatively over appointment weeks. Notably, Week 9 has the most total appointments (6.8 M), while Week 1 has the fewest (0.4 M). The graph shows a progressive decline in cumulative appointments between these extremes, providing insight into the overall trend of appointment accumulation over time.

SHEET 5 – LEAD TIME INSIGHTS:

Calculating to what degree TAs perform and how far ahead of time students schedule appointments.

1. **Slicer:** In Power BI, the slicer quickly filters data, highlighting essential insights by filtering out all TA names, hence easing analysis and focus.
2. **Pie Chart:** Illustrates the distribution of appointments by status, with status being the primary filter and slicers for TA names and lead days providing additional refinement. It provides a detailed overview of appointment status proportions, enabling customized analysis using TA and lead time factors. This image helps you understand appointment dynamics across different statuses, allowing you to spot trends and potential areas for scheduling efficiency improvements.
3. **Stacked column graph:** Depicts lead days on the x-axis and the distribution of appointment IDs on the y-axis. This image provides insights into appointment frequency relative to lead times, allowing for a better understanding of scheduling dynamics and potential correlations between lead days and appointment volumes.

SHEET 6 – LEAD HOUR INSIGHTS:

Appointments categorized by the average of lead hours and the appointment.

1. **Funnel:** Our data, which visualizes the average lead time per appointment day, indicates scheduling efficiency trends. The report, filtered by appointment day, provides insights into appointment planning and optimization.
2. **Gauge:** The report displays the number of appointment IDs that are open and booked, as well as the average lead hours. Using gauges, we visualize

the distribution of appointments based on lead time, providing actionable insights for scheduling improvement.

RECOMMENDATIONS

To: James Kinser, Indiana University

From: Deekshitha Chikkala, Keerthi Yalamanchili, Ram Swarup Bavikati

Date: May 1st, 2024

Subject: Recommendations for Evaluating TA Help Sessions

I hope this message finds you well. We have conducted an analysis of TA (Teaching Assistant) help sessions within the Kelly program, focusing on the long-term impact, student feedback, integration with course curriculum, utilization metrics, and benchmarking against peer institutions.

Below are the key recommendations based on these visualizations:

1. ***Long-term Impact Analysis:*** Evaluate the effectiveness of TA help sessions on student performance and academic accomplishment over an extended period. This analysis will provide insights into the overall efficacy of TA support within Kelly.
2. ***Student Feedback Analysis:*** Incorporate student feedback data into the evaluation process. This feedback will offer valuable insights into the effectiveness of TA assistance sessions and areas for improvement.
3. ***Integration with Course Curriculum:*** Merge TA appointment data with course curriculum data to identify specific areas or concepts where students face challenges. This integration will facilitate tailored TA assistance sessions that address specific student needs.
4. ***Utilization Metrics:*** Calculate usage metrics for TAs to assess the effectiveness of their support during help sessions. Analyzing this data will aid in optimizing TA schedules and ensuring maximum utilization.
5. ***Benchmarking Against Similar Institutions:*** Compare TA appointment data with similar institutions to identify best practices and areas for improvement. Benchmarking will enable us to adopt effective strategies from peer institutions.

Thank you for your attention to this matter.

Best regards,

Deekshitha Chikkala, Keerthi Yalamanchili, Ram Swarup Bavikati

CONCLUSION

Finally, the analysis emphasizes the significance of scheduling TA aid sessions strategically to effectively satisfy student demand. Understanding day segment patterns might help you optimize TA availability and resource allocation. The data reveals that students actively participate in TA help sessions, demonstrating a proactive approach to seeking academic guidance.

Some TAs routinely attract more appointments throughout the day, demonstrating their effectiveness in meeting students' requirements. Kelly may improve the overall effectiveness of its teaching assistant support system by detecting peak appointment hours, improving TA allocation, and using insights from TA performance, resulting in improved academic success and learning outcomes for students.